

# Cours d'urgence bilingue - 2 h

FR / EN en vis-à-vis. Lis la colonne anglaise à voix haute.

Objectif : tenir une conversation de desk en anglais madi. Pas devenir bilingue - avoir de quoi dire.

PDF pour Gemini : anglais/COURS-URGENCE-bilingue.pdf - protocole de simulation en Partie 5.

## Partie 1 - Le socle (20 min)

### 1.1 Les 15 mots

| FR                  | EN              | Prononciation                 |
|---------------------|-----------------|-------------------------------|
| obligation          | a bond          | /bond/ - pas « bonde »        |
| rendement           | the yield       | /jeeld/ - pas « yeld »        |
| coupon              | the coupon      | /koopon/                      |
| courbe des taux     | the yield curve |                               |
| point de base       | a basis point   | dit « a bip »                 |
| échéance            | maturity        |                               |
| écart               | the spread      |                               |
| volatilité          | vol             | jamais volatility sur un desk |
| acheteur            | long            | I'm long duration             |
| vendeur             | short           | I'm short gamma               |
| couvrir             | to hedge        | /hedj/                        |
| résultat            | the P&L         | dit « the P and L »           |
| sous-jacent         | the underlying  |                               |
| échéance/expiration | expiry          |                               |
| prix d'exercice     | the strike      |                               |

### 1.2 Décrire un marché

| FR                                     | EN                                |
|----------------------------------------|-----------------------------------|
| la courbe se pentifie / s'aplatit      | the curve steepens / flattens     |
| les taux montent / baissent            | yields are up / down              |
| les spreads se resserrent / s'écartent | spreads are tightening / widening |
| c'est cher                             | it's rich (jamais expensive)      |
| c'est bon marché                       | it's cheap                        |
| le marché anticipe déjà                | the market is pricing in...       |
| hausse de 25 bp                        | a twenty-five basis point hike    |
| baisse                                 | a cut                             |
| statu quo                              | the central bank held             |
| prise de bénéfices                     | profit-taking                     |
| dénouer                                | to unwind                         |

### 1.3 Le desk

| FR                  | EN                       |
|---------------------|--------------------------|
| tenir un marché     | to make a market         |
| coter à double sens | to quote a two-way price |
| écart achat-vente   | the bid-offer spread     |
| flux clients        | client flow              |

|                  |                             |
|------------------|-----------------------------|
| le portefeuille  | the book                    |
| adjudication     | an auction                  |
| pension livrée   | repo                        |
| titre recherché  | the bond is trading special |
| couvrir en delta | to delta-hedge              |
| vol implicite    | implied vol                 |

Dis ça maintenant, à voix haute :

*\*\*The ECB hiked twenty-five bips this week. The curve flattened, and the market is pricing another hike in December.\*\**

## Partie 2 - Les trois idées du cours (40 min)

### 2.1 Parité call-put · Put-call parity

FR - Acheter un call et vendre un put revient exactement à détenir l'actif à crédit. Ce n'est pas une théorie : c'est un arbitrage. À l'échéance, ce portefeuille vaut toujours  $S - K$ , quoi qu'il arrive. Deux choses qui valent la même chose demain valent la même chose aujourd'hui.

EN - *\*\*A long call plus a short put replicates a forward position on the underlying. At expiry the payoff is always  $S$  minus  $K$ , whatever happens. So the two must have the same price today - otherwise you'd lock in a riskless profit.\*\**

$$C - P = S_0 e^{-qT} - K e^{-rT}$$

Repère :  $S=100$ ,  $K=100$ ,  $r=4\%$ ,  $q=2\%$ ,  $T=0,5$ ,  $C=6,20 \rightarrow P = 5,2149$

### 2.2 Le forward · The forward

FR - Le prix forward n'est pas une prévision. C'est le spot corrigé du coût de portage : ce que coûte de détenir l'actif jusqu'à l'échéance (financement moins dividendes).

EN - *\*\*The forward is not a forecast. It's the spot adjusted for cost of carry - funding minus dividends, or the foreign rate in FX. It's a no-arbitrage price.\*\**

$$F = S_0 e^{(r-q)T}$$

$$\text{En FX : } F = S_0 e^{(rd - rf)T}$$

Réflexe : si  $r > q \rightarrow$  forward au-dessus du spot (contango).

Si  $q > r \rightarrow$  en-dessous (backwardation).

Repères : indice  $S=3500$ ,  $r=3,5\%$ ,  $q=1,8\%$ ,  $T=0,75 \rightarrow F = 3544,91$

EURUSD  $1,0850$ ,  $rd=4,25\%$ ,  $rf=2,25\%$ ,  $T=0,25 \rightarrow F = 1,090439$  (+54,4 pips)

### 2.3 Black-Scholes

FR - L'idée fondatrice est la réplication : on peut copier une option en détenant une quantité d'actif qu'on ajuste en continu. Si la copie est parfaite, l'option n'a qu'un seul prix possible.

EN - *\*\*The core idea is replication. You can copy the option by holding a quantity of the underlying and adjusting it continuously. If the replication is perfect, there's only one possible price - otherwise there's an arbitrage.\*\**

$$C = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2)$$

$$d_1 = \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

| Terme      | FR                             | EN                                      |
|------------|--------------------------------|-----------------------------------------|
| N(d2)      | proba de finir dans la monnaie | the probability of ending in the money  |
| N(d1)      | le delta                       | the delta - how much underlying to hold |
| $Ke^{-rT}$ | ce que tu paieras, actualisé   | the discounted strike                   |

Repère :  $S=100$ ,  $K=105$ ,  $r=3\%$ ,  $q=0$ ,  $\sigma=25\%$ ,  $T=0,5 \rightarrow d1=-0,102758$ ,  
 $d2=-0,279534$ ,  $C = 5,5760$ ,  $P = 9,0127$

Approximation de tête (call ATM,  $r=0$ ) :  $C \sim 0,4 \times \sigma \sqrt{T} \times S$   
 $\rightarrow \sigma=20\%$ ,  $T=1$ ,  $S=100$  : approx 8,00 vs vrai 7,9656

## Les 6 hypothèses · The six assumptions

| FR                                    | EN                                           |
|---------------------------------------|----------------------------------------------|
| 1. Mouvement brownien géométrique     | geometric Brownian motion, lognormal returns |
| 2. Volatilité constante               | constant volatility                          |
| 3. Taux constant                      | constant risk-free rate                      |
| 4. Pas de coûts ni de taxes           | no transaction costs, no taxes               |
| 5. Divisibilité, vente à découvert    | perfect divisibility, short selling allowed  |
| 6. Pas d'arbitrage, option européenne | no arbitrage, European exercise              |

La question qui tombe : « laquelle casse en premier ? »

EN - *"Constant volatility. That's exactly why the volatility smile exists - implied vol isn't flat across strikes, so the model's own assumption is contradicted by the market that uses it."*

## 2.4 Les grecs · The Greeks

| Grec  | FR                   | EN - à dire tel quel                                  |
|-------|----------------------|-------------------------------------------------------|
| Delta | sensibilité au spot  | "how much the option moves when the underlying moves" |
| Gamma | vitesse du delta     | "how fast my hedge goes stale"                        |
| Vega  | sensibilité à la vol | "my exposure to implied vol"                          |
| Theta | perte par jour       | "what I pay to hold the option overnight"             |
| Rho   | sensibilité au taux  | "rate sensitivity - usually the smallest"             |

Repères (Ex5) :  $\Delta=0,459078$  ·  $\Gamma=0,022449$  ·  $\text{vega}=0,280609$  ·  $\text{Theta/jour}=-0,022535$  ·  
 $\rho=0,201659$

### La question piège : short gamma

FR - Vendeur d'options, ton delta évolue contre toi. Tu es forcé d'acheter quand ça monte et de vendre quand ça baisse : tu achètes haut, tu vends bas.

EN - *"When you're short gamma, you're hedging into the move - you buy the highs and sell the lows. You collect theta every day, but a fast market costs you more than you collect. I saw exactly that dynamic when I back-tested the March 2020 crash on my own pricer."*

*Cette dernière phrase est ton meilleur atout. Tu l'as vécu, pas lu.*

### La relation clé

$\text{Theta} + \frac{1}{2}\sigma^2 S^2 \Gamma = 0$  (straddle,  $r = q = 0$ )

FR - Le gamma se paie en theta. EN - *"You pay for convexity with time decay. That trade-off is the whole life of an options desk."*

## Partie 3 - Parler de DB et du marché (20 min)

### 3.1 Les chiffres à connaître

| Fait              | Chiffre                                       |
|-------------------|-----------------------------------------------|
| FIC 2025          | 9,6 MdEUR, +13 %                              |
| Dernier trimestre | record : 2,6 MdEUR, +16 %                     |
| Moteurs           | Rates et Credit, +27 %                        |
| Positionnement    | #1 European FIC house                         |
| Ambition          | the only truly global alternative to US banks |
| Cible 2028        | RoTE > 14 %, C/I < 55 %                       |

### 3.2 Le marché cette semaine

| Fait                | Valeur                                 |
|---------------------|----------------------------------------|
| BCE (10/09)         | hausse de 25 bp, dépôt à 2,50 %        |
| Marché sur décembre | ~40 % de proba d'une hausse de plus    |
| OAT-Bund            | ~85 bp - haut de la fourchette d'un an |
| Inflation 2026      | 3,0 %                                  |

EN - *"The ECB hiked twenty-five bips on Wednesday, taking the deposit rate to two point five. The market is pricing roughly a forty percent chance of another hike in December. And the OAT-Bund spread is around eighty-five bips - close to the top of its one-year range, so there's a real French risk premium being priced."*

## Partie 4 - Les 3 réponses par cur (20 min)

### Tell me about yourself (~75 s)

*"I'm Calvin, a final-year Master's student at SKEMA in Paris, focusing on market finance. What got me into trading was one concrete thing: I built a lab where I model a geopolitical shock on oil and measure the P&L of a hedged book on real data. What it taught me is that timing matters more than direction - in my own numbers, exiting at the peak versus holding to the stop was worth about six hundred thousand dollars on a twenty-five million book. That's what pulled me towards a desk rather than research. I also did multi-asset reporting at BPCE Assurances, so I've worked with Bloomberg curves daily. Right now I'm going through rates and derivatives fundamentals."*

### Why Deutsche Bank? (~60 s)

*"Two reasons. First, positioning - DB is the number one European FIC house, and the strategy is explicitly to be the global alternative to US banks. Most European players are consolidating regionally; that's a different bet. Second, momentum: FIC was 9.6 billion euros in 2025, up 13 percent, and last quarter was a record - 2.6 billion, up 16 percent, driven by rates and credit. What interests me is that management says the franchise is now diversified enough not to depend on volatility. That's the harder thing to build."*

### Why rates? (~45 s)

*"Rates is the market where a view has to be expressed precisely. In equities you can be roughly right about direction. In rates you have to pick the point on the curve, the instrument, the carry. I like that the thinking is constrained - there's macro, but it has to survive contact with a curve and a repo rate. And right now it's the most alive market in Europe: the ECB hiked this week, and the OAT-Bund spread is near the top of its one-year range."*

## Les 8 phrases de secours

| Situation      | À dire                                                   |
|----------------|----------------------------------------------------------|
| Répéter        | "Sorry, could you repeat that?"                          |
| Gagner 3 s     | "That's a good question - let me think."                 |
| Tu ne sais pas | "I don't know that one yet. Can you walk me through it?" |
| Te corriger    | "Actually, let me correct myself."                       |
| Vérifier       | "Does that answer your question?"                        |
| Reformuler     | "So if I understand correctly, you're asking..."         |
| Nuancer        | "It depends on..."                                       |
| Conclure       | "That's how I'd approach it."                            |

« I don't know » dit calmement est un atout. Un junior qui invente est dangereux sur un desk. Un junior qui dit « pas encore, montrez-moi » est formable.

## Partie 5 - PROTOCOLE POUR GEMINI

Copie-colle le bloc ci-dessous à Gemini, avec le PDF en pièce jointe.

You are a Deutsche Bank rates trader conducting a 30-minute mock interview with me in English. I am a French Master's student, and English Front Office conversation is my weak point - I have never practised it.

Rules:

1. Speak only English. Ask one question at a time and wait for my answer.
2. Keep your questions short and spoken, like a real desk conversation - not written prose.
3. After each of my answers, give me three lines of feedback: one thing that worked, one thing to fix, and the better way to say it in desk English.
4. If I use French-sounding English ("expensive vol" instead of "rich vol", "volatility" instead of "vol"), correct it immediately.
5. If I freeze or stay silent, don't rescue me - prompt me with "take your time" and wait.
6. Be direct but not harsh. Trading floors are blunt, not cruel.

Run this sequence:

1. Tell me about yourself.
2. Why Deutsche Bank, not a US bank?
3. Why rates?
4. What did the ECB do this week, and what's priced for December?
5. Walk me through Black-Scholes out loud. Which assumption breaks first?
6. You're short gamma and the market gaps. What happens to you?
7. I flip a fair coin until heads. Expected number of flips? Think out loud.
8. Final one: you're on a call with a hundred people and the host asks for questions. Unmute and ask yours.

At the end, give me: a score out of 10 on fluency, a score out of 10 on technical content, and the five expressions I should drill before Tuesday. The attached PDF contains the material I've studied - use it to judge whether my answers match what I'm supposed to know. Start with question 1 now.

## Après la simulation

- [ ] Noter les 5 expressions que Gemini te donne
- [ ] Refaire les questions 1, 2 et 8 une deuxième fois
- [ ] Lundi : uniquement la question 8, dix fois

## Le test de 5 minutes - sans notes

Réponds à voix haute, en anglais.

1. What is put-call parity? Why is it true?
2. Is the forward a forecast? Why not?
3. Name three Black-Scholes assumptions. Which breaks first?
4. What does gamma measure?
5. You're short gamma and the market gaps. What happens?

1. \*A long call plus a short put replicates a forward on the underlying, so their prices are linked by arbitrage.\*

2. \*No - it's the spot adjusted for cost of carry. A no-arbitrage price, not a prediction.\*

3. \*Constant vol, constant rates, lognormal returns, no transaction costs, European exercise. Constant vol breaks first - hence the smile.\*

4. \*How fast delta changes when the underlying moves - how quickly my hedge goes stale.\*

5. \*My delta moves against me. I buy the highs and sell the lows to re-hedge. I collect theta, but a fast market costs me more than I collect.\*

*Si tu passes les 5 sans bloquer, tes bases sont là. Tu es au-dessus de la moyenne des étudiants qui seront sur le call mardi.*